

Generative Adversarial Networks (GAN)



real data



looks like real data

→ training unstable
→ hard to evaluate



Diffusion
models

i) rare dataset → training

ii) Ghibli trend

iii) Deep fakes

FNN

RNN

CNN

Transformer

GAN

GANs

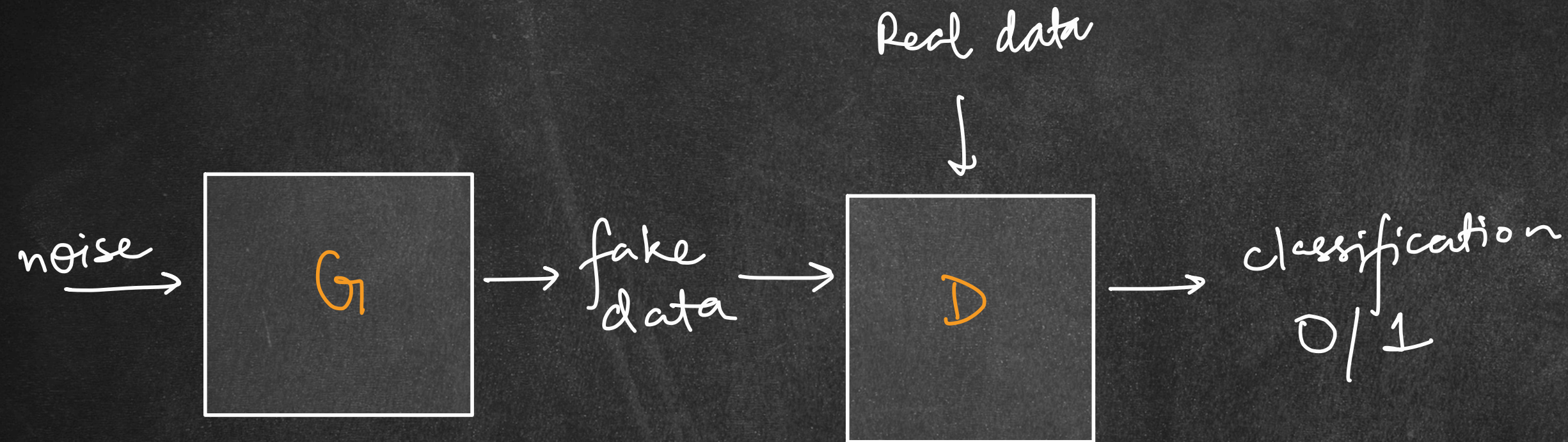
Architecture

2 NN

Generator (G)

Discriminator (D)

Generative
Adversarial
Networks



GANs

Generator Network

Vanilla GANs

$$y = Wz + b$$

fully connected (dense)

noise



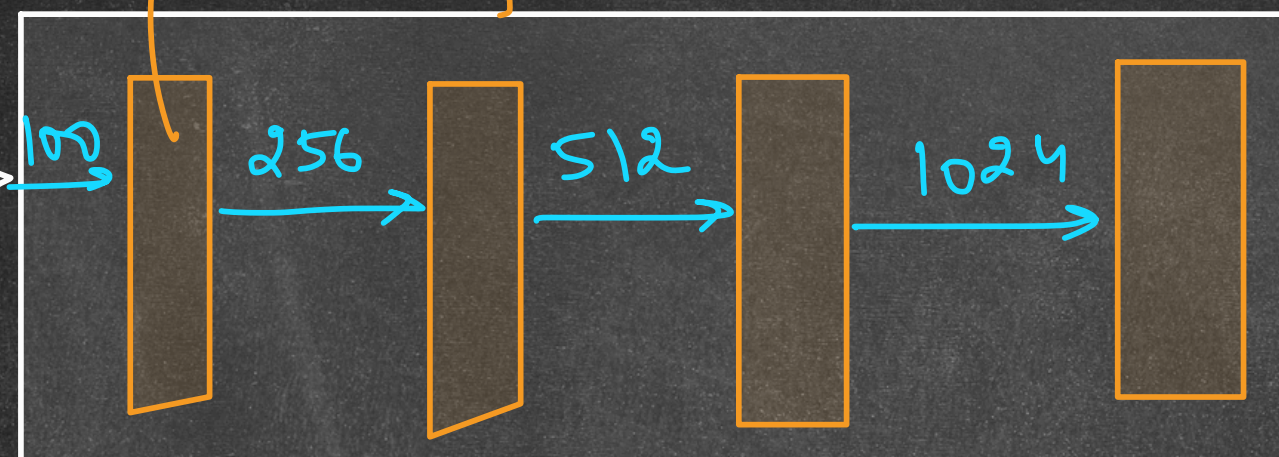
structured
fake data

noise

↓
random
noise

$[0.1, 0.32, 0.9, \dots] = z$
100

Gaussian ($\mu=0, \text{std dev}=1$)
uniform



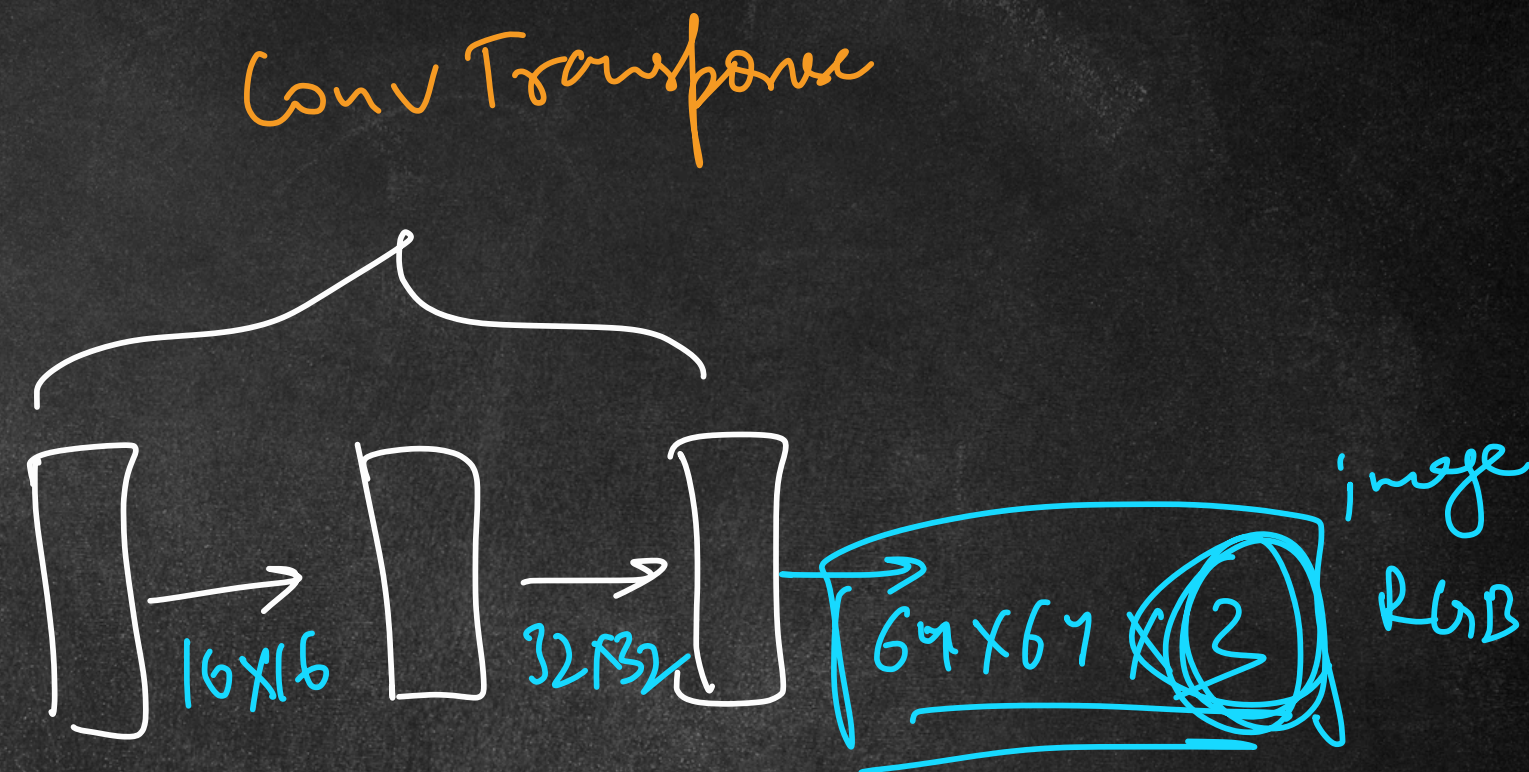
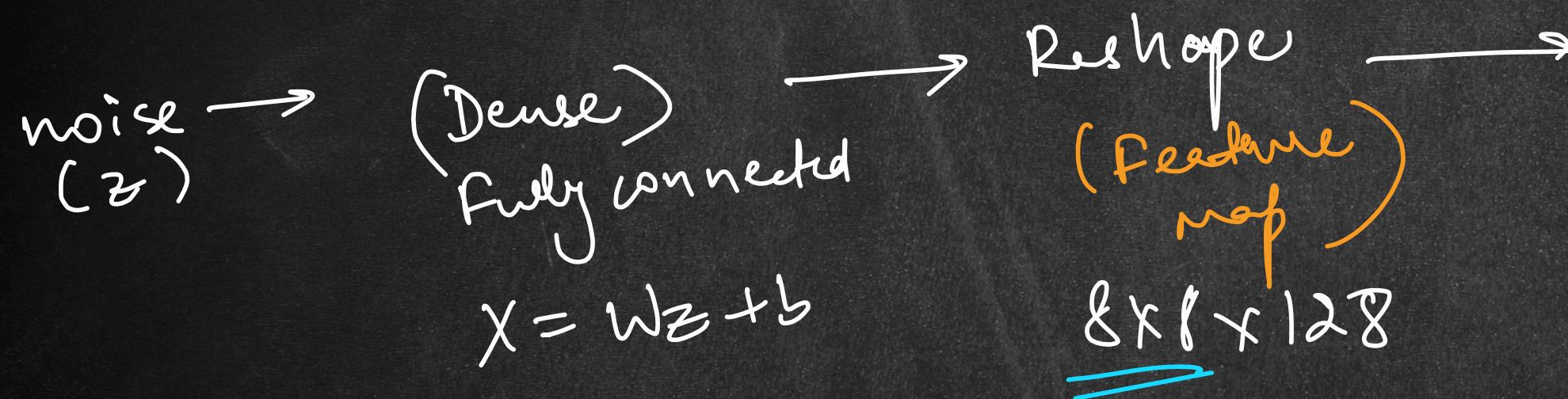
↓
UPSAMPLING (expansion)

image (RGB)
64 x 64 x 3

GANs

Generator Network

↓
DCGAN → Deep Convolutional GAN



GANs

Generator Network - Code

→ fake faces

Celeba images $\Rightarrow 178 \times 178$

square

178 x 178

GANs

Discriminator Network



① Generator loss (minimize)

$$g_loss = -\log(D(G(z)))$$

① $\approx \infty$
② $\approx 0 \checkmark$

② Discriminator loss

$$d_loss = -[\log(D(x)) + \log(1 - D(G(z)))]$$

$-\left[\log(1) + \log(1 - 0)\right] \approx 0$

GANs

Discriminator Network

→ vanilla CNN



CNN

Binary Cross Entropy

G → upsampling → generate
D → downsampling → classify

Discriminator Network - Code

GANs

Training

grid tensor $\Rightarrow$ channels $\times$ H $\times$ W
0 1 2

matplotlib $\Rightarrow$ H $\times$ W $\times$ C
1 2 0

